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 SURVEY

# Responsible AI for Cybercrime Investigations Using Knowledge-Graph and Semantic Technologies: A Systematic Literature Review

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**ABSTRACT** Cybercrime investigations in the digital space often require the analysis of evidences that are dispersed across various digital devices and computing platforms. Traditional forensic procedures often lack the necessary mechanisms and constructs to link this evidence and heterogeneous data to build formal knowledge representations for investigative tasks. Knowledge graphs and semantic technologies offer machine-readable, trackable, and formal representations of knowledge that can enable semantic reasoning and explainable artificial intelligence. In this systematic literature review, we include 43 primary studies that are published between 2020 and 2026 and synthesize the graph-based knowledge representation and semantic approaches that are applied in the cybercrime and digital forensics domains. We use narrative synthesis and quantitative analysis with descriptive summaries to integrate, analyze and further interpret the findings across multiple studies. The paper contributions demonstrate the architectural shift towards intelligent semantic digital forensic systems that use hybrid and neuro-symbolic architectures that integrate deep learning and language models and knowledge graphs. In reference to the responsible use of the AI models in such applications, we highlight the underexplored sociotechnical aspects about governance, safety, forensic standardization, bias mitigation and accountability. The results of this study can benefit scholars, practitioners, and policy-makers for future research and development of explainable and legally admissible forensic systems.

**INDEX TERMS** Cybercrime, digital forensics, knowledge graphs, responsible AI, semantic technologies, systematic literature review.

## I. INTRODUCTION

The current digital landscape is characterized by an unprecedented amount of electronic data that has emerged in different forms and is used for various purposes in intelligent computing platforms. Understanding the interconnections between these massive amounts of data and how they relate is a challenging cognitive and analytical task for analysts or human investigators [1]. However, it is significantly crucial to provide such analysis in complex or high-stake scenarios and digital environments where illegal activities, misuse, or pervasive threats may occur. In this context, cybercrime

is defined as any illegal act that is committed in cyberspace using a computer system [2].

The digital evidence related to illegal activities, fraud, or anomalous behavior can be heterogeneous and varied in scale, structure, and semantic interpretations. The evidence can be acquired from multiple sources such as network traffic logs, financial transactions, messaging platforms, cloud services, or legal records. Therefore, forensic science should provide methods to identify, analyze, and preserve the evidence for investigative cases and juridical requirements [3].

Traditional digital forensic models are largely dependent on predefined criminal investigation workflows or procedures that use isolated tools and techniques to monitor and

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analyze digital crime evidence [4]. While these methods demonstrate effectiveness in certain scenarios, they offer limited support for cross-domain investigations, semantic interoperability, knowledge representation of investigative context, and legal constraints for regulation-aware reasoning [4]. In emergent fields such as Connected Autonomous Vehicles (CAVs), a large volume of heterogeneous data from various sensors, communication interfaces, and software and hardware technologies must be analyzed for real-time incident reconstruction and investigative tasks [5]. Furthermore, the use of Opaque neural AI models for forensic tasks diminishes the interpretability and validation of investigative results and poses challenges to judicial admissibility [1], [6].

To overcome these challenges, modern crime investigation and forensic disciplines integrate semantic and formal knowledge representations (KR) and symbolic AI. Semantic technologies, including knowledge graphs, ontologies, Resource Description Framework (RDF), Web Ontology Language (OWL), SPARQL Protocol and RDF Query Language (SPARQL), Shapes Constraint Language (SHACL), and Semantic Web Rule Language (SWRL), enable machine-interpretable and trackable representations of entities, events, relationships, and constraints for explainable and interpretable AI [7], [8]. These technologies provide formal and explicit descriptions of concepts in a specific domain or discourse [9]. Furthermore, they support evidence representation normalization, integration, and reasoning across diverse sources. Ontologies have been used to operationalize semantic reasoning for investigative tasks in different sub-sectors such as event-centric forensic analysis [10], financial fraud detection [11], asset concealment [12], insider threat correlation [13], and legal reasoning [14]. For instance, in the financial investigation domain, specialized ontologies such as OntoFiC have been developed to enable domain knowledge representation and rule definition for analyzing and detecting fraud or suspicious behaviors [11]. Similarly, formal knowledge models have been used in Online Social Networks (OSNs) to provide concepts that enable the interpretation of events for investigative tasks [1].

Furthermore, recent research directions are shifting towards hybrid and neuro-symbolic architectures that combine the capabilities of neural AI with formal knowledge models. For instance, Large Language Models (LLMs) can benefit from symbolic verification to mitigate the risk of hallucination in generated responses [15]. Knowledge graphs also serve to enrich forensic evidence and provide contextual information such as historical patterns, organizational roles, and regulatory requirements to support traceable and explainable reasoning [16]. AI-assisted forensic evidence detection can benefit from forensic knowledge graphs for grounding investigative results [17].

The research and technical developments are dispersed across multiple domains such as cybersecurity, financial crime, insider threat, and legal reasoning [3].

Existing systematic reviews often highlight semantic integration gaps in digital forensic models but do not focus on graph-based knowledge representation or regulation-aware reasoning [18]. Moreover, reviews addressing the use of deep learning models such as Graph Neural Networks (GNNs) to study crime in social media or financial fraud often miss the discourse regarding semantic integration or knowledge graphs [19], [20]. Consequently, there is currently no up-to-date systematic synthesis that consolidates how semantic technologies are architected, applied, and evaluated across cybercrime domains.

The contributions of this paper include the explanation and classification of knowledge graph-based and semantic approaches, frameworks, and methods in the cybercrime and digital forensics domain [21]. Furthermore, the review maps application domains and forensic problems and identifies underexplored socio-technical and responsible AI aspects of semantic technologies and regulation-aware reasoning. The review adopts a narrative synthesis method and quantitative analysis with descriptive summaries to integrate, analyze, and interpret findings across multiple studies to answer the research questions. It benefits researchers, practitioners, and policymakers by providing deeper insight into recent methods, domain-specific use cases, challenges, and governance implications.

## II. RESEARCH METHOD

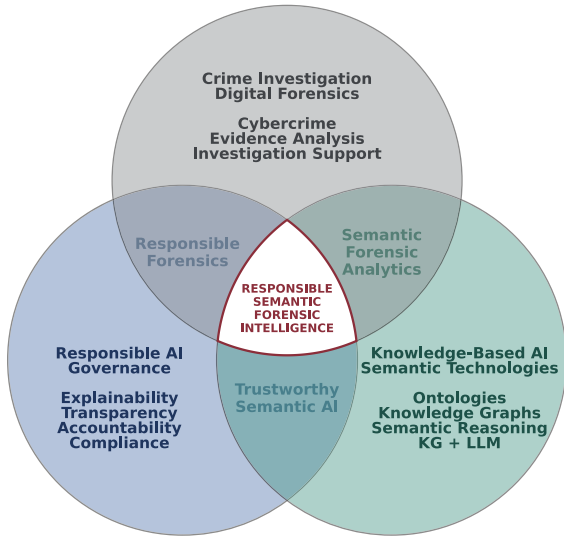
This systematic literature review (SLR) is structured to identify extant research and examine recent graph-based knowledge representation and semantic technologies that are used in forensic analysis and crime investigation. We adhere to the Kitchenham methodology for organizing different steps of the review and to ensure rigor and reproducibility of the results [22]. The review is conceptually positioned in the intersection of Responsible AI Governance, Knowledge-Based AI and Semantic Technologies, and Crime Investigation and Digital Forensics as illustrated in Fig. 1.

### A. RESEARCH QUESTIONS

The primary goal of this review is to synthesize recent architectures and approaches in research on how semantic technologies and knowledge graphs are used for crime investigation. We review how different domains use knowledge representation structures and semantic technologies for digital investigations. Furthermore, socio-technical and responsible AI aspects and challenges of using these technologies are addressed. The research is guided by three main research questions (RQs):

- **RQ1:** What are the recent approaches and architectures in graph-based knowledge representation and semantic technologies applied to crime investigation, fraud detection, and forensics?
- **RQ2:** In which domains (e.g., cybersecurity, financial fraud, legal reasoning) and for which investigative problems have these technologies been applied?

- **RQ3:** Which sociotechnical and responsible AI aspects of knowledge graphs and semantic technologies for cybercrime remain underexplored (e.g., accountability, explainability, bias, risk, governance, ethics, etc.)?



**FIGURE 1.** Research scope and contribution of this systematic literature review.

## B. SEARCH STRATEGY

An automated search strategy is employed as the primary approach and is complemented by backward snowballing to capture influential primary studies that are not retrieved by the initial queries. Backward snowballing is applied in one iteration to all included studies from database searches that pass quality assessment criteria. The resulting papers are further assessed using the same inclusion and exclusion criteria and quality rules. This strategy ensures both breadth and depth of coverage in a rapidly evolving research area. The concrete search queries are saved for each database along with the search date to enable refreshing the list of primary studies. The results of the automated search include 597 papers from six major databases, as shown in Table 1. There are 120 duplicate papers detected in Covidence and five duplicates manually detected, resulting in 472 non-duplicated primary studies.

**TABLE 1.** Number of papers by database.

| Database / Library               | # of Papers |
|----------------------------------|-------------|
| Scopus                           | 309         |
| Web of Science (Core Collection) | 117         |
| IEEE Xplore                      | 118         |
| ACM Digital Library              | 21          |
| ScienceDirect                    | 2           |
| SpringerLink                     | 30          |
| Total                            | 597         |
| Total (deduplicated)             | 472         |

## 1) SOURCE DATABASES

The search was conducted across six major digital libraries and databases to provide comprehensive coverage of

computer science, cybersecurity, and the Semantic Web. These databases include Scopus, Web of Science Core Collection, IEEE Xplore, ACM Digital Library, SpringerLink, and ScienceDirect. Scopus and Web of Science were selected for their broad and complementary coverage of peer-reviewed literature, with Scopus emphasizing journals and conference articles and Web of Science providing stronger coverage of foundational studies. Furthermore, IEEE Xplore and the ACM Digital Library were selected to cover domain-specific conference and workshop papers. SpringerLink and ScienceDirect provide access to newer papers that are not yet indexed in Web of Science or Scopus due to selection and metadata delays. Searches were customized and adapted to each database’s syntax and applied to titles, abstracts, and keywords, and results were merged and de-duplicated in Covidence prior to screening.

## 2) QUERY FORMULATION

The search string was constructed using four conceptual clauses designed to capture the intersection of symbolic AI paradigms and investigative domains, as shown in Table 2. These clauses were combined using the Boolean AND operator. Furthermore, a detailed query for each search engine is included in APPENDIX.

**TABLE 2.** Conceptual groups used in the search query.

| Concept                            | Search Terms                                                                                                                                                                                                                                                                         |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Semantic/Symbolic Paradigm         | (semantic* OR symbolic* OR neurosymbolic)                                                                                                                                                                                                                                            |
| Knowledge Representation Structure | (knowledge-graph* OR "knowledge graph*" OR KG OR SPARQL OR RDF OR OWL OR "linked data" OR ontolog* OR SHACL OR SWRL)                                                                                                                                                                 |
| Investigative Domain               | (crim* OR fraud* OR forensic* OR cyber*crim*)                                                                                                                                                                                                                                        |
| Responsible AI Aspects             | (transpare* OR explaina* OR ethic* OR moral* OR responsibl* OR socio* OR policy* OR bias* OR human* OR value-cent* OR value-sensi* OR fair* OR accountab* OR interpret* OR governance OR audit* OR regulat* OR safe* OR trust* OR risk* OR privacy OR dignity OR agency OR autonomy) |

These clauses align with the research questions and focus on graph-based knowledge representation mechanisms, cybercrime application domains and investigative problems, and the socio-technical dimensions of semantic technologies. The inclusion of specific standards such as SHACL and SWRL ensures that the search captures studies involving formal reasoning and validation in forensic contexts. Furthermore, the temporal scope of the search is restricted to articles published from 2020 onward.

### 3) SUPPLEMENTAL SNOWBALLING

To capture influential studies that might use non-standardized terminology such as ontology-driven reasoning without explicit mention of knowledge graphs, one iteration of backward snowballing was applied to all studies passing the quality assessment criteria. This involved manually reviewing reference lists to identify influential works not retrieved by the initial Boolean queries.

## C. SELECTION CRITERIA

Selection of primary studies was conducted through a multi-stage filtering process based on predefined inclusion and exclusion criteria to maintain scientific validity.

### 1) INCLUSION CRITERIA

- Propose, utilize, or evaluate knowledge-based AI incorporating graph-based models such as Knowledge Graphs (KG), RDF, or SPARQL.
- Implement formal ontologies (e.g., OWL) or semantic reasoning rules (e.g., SWRL, SHACL) in criminal or fraud investigation contexts.
- Published as peer-reviewed journal articles or conference papers between 2020 and 2026.

### 2) EXCLUSION CRITERIA

- Non-peer-reviewed publications, book chapters, or studies not written in English.
- Use of the term “semantics” in a purely statistical or feature-based sense without formal semantic modeling.
- Rule-based systems or machine learning models that do not employ graph-based or ontological knowledge representation (KR).
- Papers not focused on cybercrime, fraud, forensics, or criminal investigations.
- Publications from predatory journals.

The publication date, peer-review status, and language criteria were applied during title and abstract screening in Covidence, resulting in the exclusion of 311 out of 472 non-duplicated papers. The remaining 161 papers were examined during full-text review. Forty-five articles were excluded for not focusing primarily on cybercrime, fraud, forensics, or criminal investigations, and five were excluded due to the absence of formal semantic modeling. Additionally, 11 irrelevant papers were excluded during full-text screening. In total, 104 articles remained, including snowballed papers. Two papers could not be retrieved from their sources, leaving 102 papers for quality assessment.

## D. QUALITY ASSESSMENT (QA)

Candidate papers were evaluated against four quality criteria (QC) on a 0–2 scale, where 0 indicates absence, 1 partial fulfillment, and 2 strong evidence:

- **QC1:** Clarity of research objectives, scope, and contributions.
- **QC2:** Methodological rigor and soundness.

- **QC3:** Evidence of validation.
- **QC4:** Substantive discussion of limitations and responsible AI aspects.

Studies with a cumulative score of at least 4 out of 8 and a QC4 score of 2 were selected. After applying these criteria, 43 papers remained for data extraction. Fig. 2 demonstrates the number of quality assessed conference and journal publications over time since 2020.

## E. DATA EXTRACTION AND SYNTHESIS

Data extraction used a structured form capturing knowledge representation (KR) types (e.g., knowledge graphs, Ontology, RDF, OWL), knowledge retrieval, logic formalisms (SWRL, FOPL, DL, etc.) and validation. Investigative issues and domains as well as socio-technical aspects were extracted. Evidence synthesis employed two methods:

- 1) **Narrative Synthesis:** Used for RQ1 and RQ3 to classify architectural trends and socio-technical challenges. Studies were grouped by KR types and cross-study comparison of symbolic and neuro-symbolic/hybrid architectures. Furthermore, the machine learning methods used within each architectural paradigm is demonstrated over time. Additionally, for RQ3, socio-technical and responsible AI aspects such as accountability, governance, and bias mitigation were thematically analyzed based on extracted responsible AI description to find underexplored categories.
- 2) **Domain-Oriented Comparative Analysis:** Used for RQ2 to map investigative tasks across application domains [1], [23].

The study venues (conference vs journal) over time were demonstrated. The review also maps evidence of knowledge graph and symbolic technologies and their conceptual subcategories across primary studies and includes trend analysis and cross-study comparison of symbolic and neuro-symbolic/hybrid architectures. Furthermore, the machine learning methods used within each architectural paradigm is demonstrated over time. Additionally, for RQ3, socio-technical and responsible AI aspects were thematically analyzed based on challenge description in data extraction form to find underexplored themes and responsible AI aspects.

temporal trend analysis (2020–2026) to track the technology adoption of specific semantic standards (e.g., RDF, OWL, SWRL) and architectural classification to identify the decisive paradigm shift from purely symbolic models to hybrid neuro-symbolic systems

**Evidence Mapping and Frequency Analysis:** State that in addition to basic frequency-based counts, the review employs evidence mapping to visualize research density across intersecting domains (e.g., Finance vs. Cybersecurity) and investigative problems. **Longitudinal Technology Adoption and Temporal Trends:** Explicitly mention that the synthesis includes temporal trend analysis (2020–2026) to track the adoption rate of specific semantic standards like RDF, OWL,

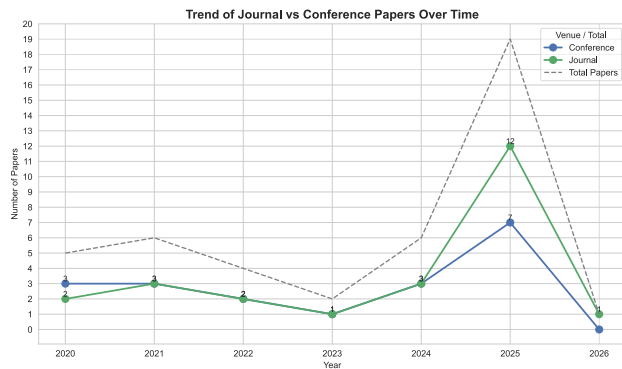


FIGURE 2. Trend of Journal vs Conference papers over time.

and SWRL. Architectural Transition Analysis: Detail the methodology for architectural classification, which identifies the decisive shift from isolated Symbolic systems (Logic + Ontologies) to Hybrid Neuro-Symbolic architectures (Neural Learning + Symbolic Reasoning). ML Category Mapping: Note that the synthesis maps specific machine learning models (e.g., GNNs, RNNs, LLMs) to their functional roles within the forensic pipeline (e.g., extraction, refinement, or classification)

### III. DATA EXTRACTION AND SYNTHESIS RESULTS

In this section, we synthesize the findings based on the 43 selected primary studies and address the research questions. The paper venues over time are shown in Fig. 2. The PRISMA document is shown in Fig. 3. The results are grounded in the data that are extracted in a structured form which captures knowledge graphs and various semantic technologies for RQ1. Furthermore, RQ2 covers the application domains and the investigative problems in which the knowledge graph and semantic technologies are used. Finally, for RQ3, the sociotechnical and responsible AI aspects and challenges are discussed based on the evidence from these papers.

#### A. RQ1: RECENT APPROACHES AND ARCHITECTURES IN GRAPH-BASED KNOWLEDGE REPRESENTATION

The literature reveals that there is an urge in law enforcement agencies to adopt new technical infrastructures and architectures for crime investigation. Systems that can represent and infer knowledge are suitable for finding motifs and hidden relations that are present in data and for constructing meaningful information. Therefore, they are more suitable for crime analysis in law enforcement agencies (LEAs) compared to traditional relational database systems or card files that lack this capacity [24].

AI systems that incorporate explicit knowledge representations and reasoning mechanisms provide human-understandable explanations for their outcomes. Symbolic AI systems use formal logic and graph-based structures. A key component of these graph-based structures is ontologies,

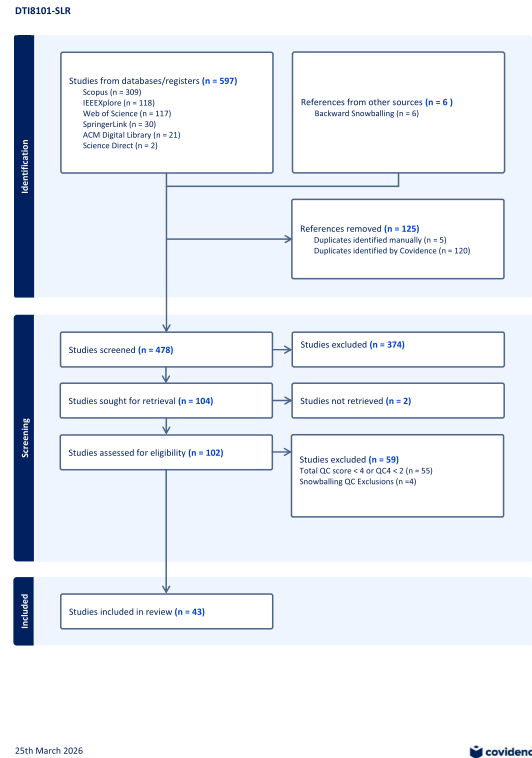


FIGURE 3. PRISMA flow diagram of study selection.

which are standard vocabularies that allow researchers to annotate, share, and reuse knowledge in various disciplines [25]. Furthermore, knowledge graphs are data models that capture real-world knowledge and consist of nodes and edges that represent entities and their relationships. These knowledge representation mechanisms are used at the core of symbolic AI for reasoning. Furthermore, neuro-symbolic AI systems are hybrid and combine semantic rules and robust learning features of neural networks to improve interpretability in neural AI systems [15].

#### 1) KNOWLEDGE REPRESENTATION

In this section, we study the results of knowledge representation that include knowledge graphs and RDF/OWL. Articles [15], [24], [26], [27], [28], [29], [30], [31] commonly use knowledge graphs and RDF/OWL ontologies.

##### a: KNOWLEDGE GRAPH

In this review, 22 papers employ knowledge graphs for knowledge representation, as demonstrated in Table 3. Knowledge graphs are structured and organized logical representations of entities and their relationships that are well suited for domain-specific knowledge [27], [30], [32]. These graph-based structures can capture and represent real-world knowledge and semantic and contextual connections and represent them by nodes (entities) and edges (relationships) [15], [33]. The relationships in graphs enable verifiable digital evidence analysis between entities and can be directed [7], [16], [17]. These graph structures can be

used with specialized algorithms for multi-hop reasoning and inference [31], [34]. Stacked Autoencoders (SAEs) can generate embeddings from knowledge graphs for reasoning tasks [35]. Liu et al. [36] use KGs to encapsulate the behavioral patterns of attackers. Papers [6], [24], [28], [29], [37], [38], [39] construct or automatically extract semantically meaningful or layered knowledge graphs from RDF triplets or ontologies. Cai et al. [40] construct a KG with semantic and syntactic layers.

#### *b: RDF / OWL / LINKED DATA*

There are 23 studies in this review that use the Resource Description Framework (RDF) and Web Ontology Language (OWL) to represent knowledge, as shown in Table 3. RDF is a fundamental Semantic Web data model that stores information and formally represents knowledge in the form of triples ( $s, p, o$ ), where  $s$  is the subject,  $p$  is the predicate, and  $o$  is the object. RDF statements can represent labeled directed multigraphs [28], [41]. Web Ontology Language (OWL) is an ontology description language based on Description Logic (DL) and a semantic extension of RDF and RDF Schema that represents and constructs ontology models [23], [42], [43], [44]. Ontology-based models and approaches provide a structured form of knowledge representation in dynamic environments [5], [45]. RDF triples can intrinsically form labeled directed multigraphs and formal representations of concepts and properties in the form of ontologies and enable integrity checking and an integrated view of entities and their relations [41], [46]. Furthermore, OWL-based reasoners can process data represented in RDF and infer rules and relevant knowledge [47], [48].

#### 2) KNOWLEDGE RETRIEVAL

SPARQL Protocol and RDF Query Language (SPARQL) is a standard set of specifications that provide languages and protocols to query and manipulate RDF data stores [23], [42]. Twelve studies in this review use this standard. Abrouk et al. [11] use it to visualize transactions through graphs to identify fraudulent transactions. Ahmed et al. [45] use SPARQL to retrieve implicit facts after the inference engine infers relevant fraud information. Similarly, Minardi et al. [46] use SPARQL to obtain point-of-interest (POI) risk information based on rules and properties that are encoded in the ontology. These queries have clauses similar to SQL and enable querying semi-structured data or metadata from local or remote heterogeneous sources [23], [48]. Arshad et al. [49] use semantic queries to extract normalized RDF stores across multiple social media platforms. SPARQL is also used to evaluate ontologies and in evaluator agents to monitor crime scenes, verify suspect information, and validate the chain of custody of digital evidence [44], [50]. Sikos [41] uses SPARQL to verify malicious network communication hypotheses. Saluja et al. [31] translate natural language queries in LLMs into equivalent SPARQL queries that can run on ontology-backed evidence bases.

Retrieval-augmented generation (RAG) methods have been used alongside LLMs to retrieve external domain-specific knowledge to address limitations such as hallucinatory response generation without retraining models [15]. Knowledge-Augmented Generation (KAG) provides access to structured information such as knowledge graphs, topology, and role-based relations and enables better context-sensitive knowledge retrieval, in contrast to RAG, which may fail to preserve essential structural semantics [6], [15].

#### 3) REASONING AND LOGIC FORMALISM

Semantic Web Rule Language (SWRL) is a standard rule language of the Semantic Web and is used to express rules and logic using antecedent and consequent forms. It can increase the expressiveness of OWL and couple it with Horn-like rules and Description Logic programs to deduce implicit knowledge from explicit facts [35], [42], [47]. Eleven primary studies in this review use SWRL, as listed in Table 3. Abou-Rida et al. [5] implement SWRL rules that are aligned with the MITRE ATT&CK cybersecurity framework using SWRLAPI in Java and extend OWL ontologies. This provides rule-based reasoning and traceable inference about detected cyberattacks or anomalies. Jabardi and Hadi [47] detect fake social media accounts using interpretable SWRL decision rules. Abrouk et al. [11] use a rule-based approach in which SWRL rules are associated with fraud scenarios and patterns. SWRL rules can be inferred using reasoners such as Pellet to construct new links between entities in ontologies and deduce new knowledge in real time. The inferred knowledge can be stored back as new triplet facts [23], [24]. SWRL rules can also be used for data correlation, logic, and proof for validation [41]. Chen et al. [43] incorporate SWRL rules by extending the ontology model with 35 rules to support effective reasoning conclusions. Arshad et al. [1] emphasize that automated forensic processes require formal theories and use predicate logic concepts to formalize knowledge. Article [40] uses AnyBURL, a knowledge graph completion method based on logical rules.

#### 4) VALIDATION AND CONSTRAINTS

Saluja et al. [31] cross-validate the generated RDF triplets with the rules and constraints of the domain ontology using the Shapes Constraint Language (SHACL) to exclude anomalies.

Fig. 4 shows a heatmap of the use of semantic categories and knowledge graphs over time for each primary study. The reviews by [51] and [52] are not depicted in Fig. 4 since they provide broader studies of various explainable methods. Fig. 5 shows that Knowledge Graph (KG) is the category with the highest number of papers.

#### 5) SYMBOLIC ARCHITECTURES

Symbolic ontology-based investigative systems can use standardized ontology-based semantic technologies that

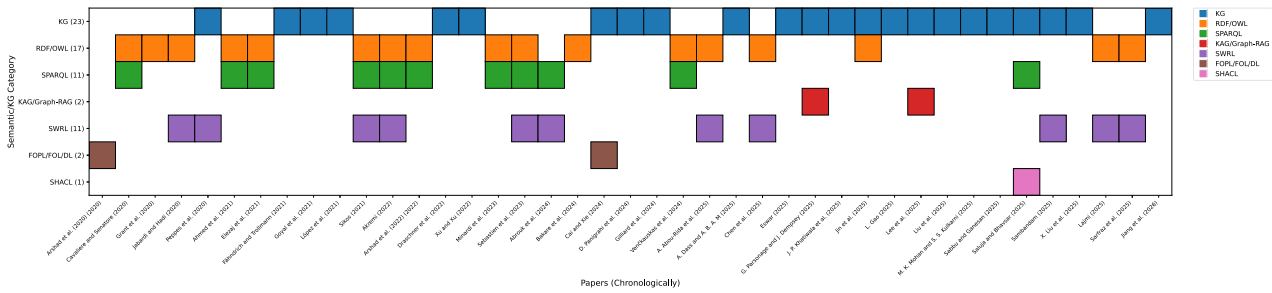


FIGURE 4. Systematic mapping of papers vs Knowledge graph / Semantic technologies.

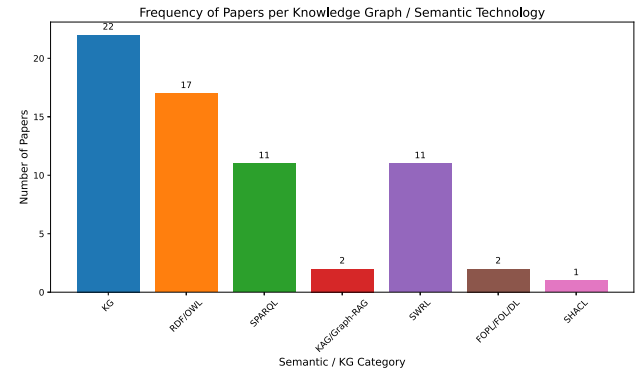


FIGURE 5. Frequency of papers per Knowledge graph / Semantic technology.

provide structured and expressive knowledge representation and retrieval and enable semantic reasoning with logic formalisms [5].

a: RULE-BASED FORENSIC INFERENCE

Abou-Rida et al. [5] employ Semantic Web Rule Language (SWRL) over ontologies to provide traceable inference and consistent semantic monitoring for detected cybersecurity attacks. Akreimi [23] infers SWRL rules with the Pellet reasoner, which create links between events and deduce new knowledge about committed fraud. Grant et al. [25] present the implementation of SWRL rules and use the PELLET reasoner to fire rules against ontologies for forensic event analysis. Ahmed et al. [45] propose an alert model that infers facts from ontologies and developed rules (Apache Jena Reasoner) that are used by the inference engine for detecting suspicious transactions. Article [43] also uses predefined SWRL rules that support ontology reasoning in the Pellet reasoner to detect misstatement accounts and financial fraud and generate risk warnings. Jabardi and Hadi [47] utilize an ontology classifier that provides insight into process decisions derived by a predefined rule-based (SWRL) reasoner for fake social media profiles. Sébastien et al. [42] use ontologies coupled with Description Logic rules to deduce implicit knowledge from explicit facts for symbolic reasoning.

TABLE 3. Knowledge graph and semantic technologies.

| Category            | Instance             | Papers (Author and Year)                                                                                         | Freq. |
|---------------------|----------------------|------------------------------------------------------------------------------------------------------------------|-------|
| KR Type             | Knowledge Graph (KG) | [32] [27] [30] [15] [33] [17] [16] [7] [34] [31] [36] [37] [29] [28] [38] [24] [6] [39] [40] [35] [26] [53] [54] | 22    |
|                     | Ontology/RDF / OWL   | [28] [42] [43] [23] [44] [5] [45] [41] [46] [47] [48] [49] [50] [25] [55] [9] [56]                               | 17    |
| Knowledge Retrieval | SPARQL               | [23] [42] [11] [45] [46] [23] [48] [49] [50] [44] [41] [31]                                                      | 11    |
|                     | KAG / Graph-RAG      | [15] [6]                                                                                                         | 2     |
| Reasoning / Logic   | SWRL rules           | [5] [11] [23] [43] [24] [35] [41] [47] [42] [55] [56]                                                            | 11    |
|                     | FOPL / FOL / DL      | [1] AnyBurl (FOL) [40]                                                                                           | 2     |
| Validation          | SHACL                | [31]                                                                                                             | 1     |

b: QUERY-DRIVEN KNOWLEDGE DISCOVERY

In semantic architectures, RDF/OWL structures can be queried to retrieve data about original ontology facts or inferred knowledge. Arshad et al. [1] propose a multi-layer approach that helps investigators pose high-level SPARQL queries to extract online social network data from RDF triples. Elezaj et al. [44] propose a semantic tool (CISMO) that translates competency questions into SPARQL queries to validate ontologies for online social network crimes. Minardi et al. [46] employ predefined SPARQL queries implemented in Apache Jena to generate relevant and novel risk mini-models from ontological relationships for crime analysis at points of interest (POIs) in smart cities. Elezaj et al. [44] build a human factor ontology based on socio-cognitive characteristics and execute SPARQL queries to evaluate the ontology and detect suspect evidence from Online Social Networks (OSNs).

c: FORMAL KNOWLEDGE MODELING

Arshad et al. [1] build an event-based formal knowledge model and ontology from heterogeneous sources that help understand evidence for forensic analysis in online social networks.

6) HYBRID AND NEURO-SYMBOLIC ARCHITECTURES

a: KG-GUIDED INTEGRATION

These architectures employ recent approaches that integrate deep learning or large language models (LLMs) with Knowledge Graphs (KGs) for validation or hallucination mitigation.

Article [40] uses a two-layer knowledge graph (FSFD-TLKG) with clear semantics for mining fraudulent patterns in financial statements. Patterns are extracted from knowledge

graph layers and used to generate a rule vector to explain classification results and improve detection performance. Gao [34] introduces a framework that uses knowledge graphs and GNNs for financial fraud detection, in which analytical tasks are defined as graph traversal problems. Beyond Budgeting Round Table (BBRT) principles are used as constraints that guide multi-hop graph traversal and path finding.

Gilliard et al. [27] use TransH knowledge graph embeddings to enable reasoning in machine learning models for network attack recognition from system state data. The embeddings are optimized by loss functions during training. Panigrahi et al. [26] use GNNs trained on an Intrusion Response System Knowledge Graph (IRSKG) of system logs and rules of engagement (RoEs) to recover compromised systems autonomously. Sabbu and Ganesan [33] use a separate semantic layer with a contextual knowledge graph combined with deep learning models such as LSTM, CNN, and RNN to recognize unusual human behavior.

Jin et al. [28] construct RDF triples mapped from network traffic to build an intrusion detection knowledge graph named IDS-IKGA, which is integrated with LightGBM and Random Forest (RF). Goyal et al. [38] use a hybrid knowledge graph-based technique integrated with Sentence-BERT (S-BERT) to incorporate contextual, factual, and meta-features to validate the plausibility of job postings. Liu et al. [36] use a multi-stage neuro-symbolic pipeline with RNN, BERT, and CNN models that leverage a user preference vector based on knowledge graphs to predict attack behaviors. Mohan and Kulkarni [30] analyze knowledge graph structures using Rate of Variable Change (RVC) analysis and embeddings learned in an autoencoder (AE) to improve financial fraud detection. Fähndrich et al. [39] use marker-passing algorithms for symbolic reasoning in knowledge graph structures and propose marker interpretations to simulate connectionist approaches such as ANNs in digital forensics.

Article [32] combines an LLM powered by the Yi-34B model with a lightweight knowledge graph layer to apply rule-based fraud keyword mappings. This reinforces domain-specific predictions and fixes incorrect financial complaint classifications caused by baseline LLMs. Jiang et al. [37] use KG expressions optimized in a graph attention network module to verify the rationale of LLM-generated text descriptions (Claude 3.5 Sonnet) of video frames for violence detection during the reasoning process.

Khatiwala et al. [17] use Digital Forensic Knowledge Graph (DFKG) visualization and cross-referencing based on deterministic hashes for structured and verifiable analysis to mitigate bias in LLM-extracted artifacts and enhance forensic authenticity and investigative reliability.

Lee et al. [6] use a RAG framework (DF-GRAPH) to encode message relations, speaker roles, and temporal links in a knowledge graph. The integration of discourse-aware graphs and rule-based trace generation guides LLMs (GPT-4o) for evidence-grounded forensic question answering. This integration achieves higher fidelity, traceability, and

legal justification. Finally, Saluja and Bhavsar [31] use a layered hybrid architecture for cloud forensic investigations that combines LLMs at the front end with ontology-backed forensic knowledge graphs that provide evidence mapping and temporal correlations. X. Liu et al. [54] integrate knowledge graphs with a gated fusion mechanism to adaptively weight RoBERTa (LLM) and GNN embeddings and uncover complex fraud relationships and semantic patterns.

#### *b: ONTOLOGY-GUIDED INTEGRATION*

Parsonage and Dempsey [15] use a neuro-symbolic approach that combines bottom-up learning from data with top-down reasoning. Forensic meta-information is captured in ontologies for KAG training and transformed into Answer Set Programming (ASP) rules. Furthermore, RDF triples generated by multi-modal LLMs (MM-LLMs) are validated against ASP rules. López et al. use dependency parsing and fine-tuned BERT-based text classifiers guided by regulatory healthcare policy ontologies and convert them into human-understandable and machine-executable benefit rules.

Eswar [53] proposes a real-time neuro-symbolic self-explanatory AI (NS-XAI) framework that combines deep learning MLPs with symbolic AI to generate human-readable explanations. NS-XAI provides bias control and does not require additional post-processing for explanations, unlike post-hoc techniques such as SHapley Additive exPlanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME). Sambandam [35] combines ontologies and domain-specific knowledge bases with semantic rules and Doc2Vec to transform concepts and rules into multidimensional vector embeddings. Neural and deep learning models such as SAEs (Stacked Autoencoders) and Deep Neural Networks (DNNs) use these embeddings for training and classifying normal and abnormal transactions. Peppes et al. [24] use a framework with a semantic layer based on OWL ontologies and knowledge graphs to provide a common communication language. ANNs and Markov Logic Networks (MLNs) leverage the structures, entities, and relationships in this language for crime analysis.

### 7) MODULARITY

#### *a: INDEPENDENT STAGES AND STRUCTURAL HIERARCHIES*

Articles [27], [42], [47] feature multiple methodological stages, steps, or flows for specific tasks such as knowledge acquisition, domain modeling, ontology building, rule mining, and reasoning. Grant et al. [25] use slice-based hierarchies to manage ontology components at different levels. Elezaj et al. [44] and Sambandam et al. [35] discuss ontology metadata mapping with NLP, Doc2Vec, and deep learning components. These multi-stage or hierarchical systems are categorized as partially modular systems in this review as shown in Table. 4.

#### *b: MULTI-LAYERED PLUGGABLE ARCHITECTURES*

Saluja et al. [31] propose an architecture with six separate pluggable layers that include data source, preprocessing,

modeling, visualization, application, and data provenance. The architecture in [45] consists of three specific layers for domain conceptualization, ontology, and rules. The architecture in [53] uses neural and symbolic components and integrates them with a self-explanatory layer for providing explanations. Arshad et al. [1] use five independent layers for knowledge, process, data, analysis, and interface, which are implemented using different models and techniques at each layer for digital forensics in online social networks. We categorize these multi-layer pluggable models as mostly modular systems.

### c: MULTI-AGENT AND SCALABLE PIPELINES

Cavaliere and Senatore [50] create a multi-agent-based design with four types of agents—Collector, Contextualizer, Reasoning, and Evaluator—for intelligent monitoring. The agents use direct interaction mechanisms and broadcast signals to communicate results when tasks are accomplished. Khatiwala et al. [17] use a scalable pipeline for data extraction, LLM refinement, and graph-based correlation. Similarly, L. Gao uses a scalable pipeline for data acquisition, query processing components, and a reasoning engine. Dass et al. [32] employ separate and parallel pipelines for fraudulent or KG-based false complaint classification and detection to process datasets. These agentic systems and scalable pipelines are categorized as mostly modular.

### d: FULLY MODULAR INDEPENDENT CLUSTERS

Minardi et al. [46] propose an architectural framework with separate modules for GeoQuery, semantic query, semantic risk generator, and integration with the OpenStreetMap (OSM) module in a knowledge base and GIS interface to assess crime risks. X. Liu et al. [54] use graph-representation learning, discriminant classification modules, and a dynamic weighting module to weight semantic and structural embeddings. Akremi [23] uses a cluster-based architecture consisting of application, management, front-end, and back-end self-contained clusters that process data locally and share only relevant information between checkpoints. Lee et al. [6] define a modular graph-based digital forensic system (DF-GRAPH) with distinct knowledge representation, retrieval, and LLM-based response generation components. Abou-Rida et al. [5] use a modular system that applies information management to model heterogeneous data and cyber-forensic modules that apply SWRL rules for the investigation of cyberattacks.

## 8) DISCUSSION OF RQ1: GRAPH-BASED KNOWLEDGE REPRESENTATION AND SEMANTIC ARCHITECTURES

### a: KEY FINDINGS AND COMPARATIVE ANALYSIS

The systematic inquiry in this review allowed to identify and analyze graph-based knowledge representation and various semantic technologies in symbolic and neurosymbolic architectures at different level of modularity. RQ1 specifically addresses recent approaches and architectures

in graph-based knowledge representation and explains various semantic technologies that are widely used for forensic and crime investigation. The quantitative results demonstrate that knowledge graphs (KGs) are adopted in 22 studies to represent domain-specific entities and their logical relationships and enable multi-hop reasoning over these structures to discover crime patterns [27], [30], [34], [36]. Furthermore, seventeen studies utilize the Resource Description Framework (RDF) and Web Ontology Language (OWL) to store information using triples to enable formal grounding and integrity checking [41], [43]. Semantic Rule-based logic is also predominately used to implement the reasoning capabilities. Specifically, Semantic Web Rule Language (SWRL) is used to define antecedent-consequent rules to deduce implicit knowledge from explicit facts in various investigative systems [23], [43], [47]. To ensure the validity and correctness of these inferences, Shapes Constraint Language (SHACL) is used in a study to cross-validate RDF triplets against domain-specific rules and exclude anomalies from investigative evidence [31]. The use of knowledge graphs and semantic technologies such as RDF, SPARQL, SWRL and SHACL signifies the role of knowledge synthesis in crime investigation tasks. The literature shows a transition toward architectures and methods that enable LEAs to formalize and infer hidden knowledge from diverse sources compared to relational databases or card files [24], [45]. This methodological shift is also realized in various decentralized system designs: semantic layering, multi-agent pipelines, and independent clusters are prevalent approaches to improve the modularity of investigative systems that use semantic technologies and knowledge graphs [6], [27], [31], [46], [47]. Architecturally, these multi-layered and modular designs (e.g., KDCFI, ForenSys) enable the separation of data ingestion and semantic reasoning, which is critical for maintaining criminal evidence data provenance while scaling automated analysis tasks [5], [31]. Furthermore, frameworks such as, Knowledge-Augmented Generation (KAG) and Retrieval-Augmented Generation (RAG) ground generative AI in knowledge graphs to mitigate hallucination, which provides a traceable and legally defensible reasoning that is essential for admissibility in the court of law [6], [15], [16]. Traditionally, the crime investigation approaches either favored statistical data mining or rigid expert systems where symbolic AI paradigms utilize formal logic and graph-based structures such as ontologies to share and reuse knowledge [1], [25]. The recent publications, however, integrate these approaches to create neuro-symbolic and hybrid architectures that combine robust machine learning and deep learning models with semantic rules to improve the interpretability [15], [40]. Today, hybrid and neuro-symbolic methods are commonly used approaches that employ a variety of classical machine learning, deep learning, semantic graphs, knowledge graph embeddings, GNNs, and LLMs as shown in Fig. 6 and Fig. 7. Reviews in [51] and [52] are not reflected in Fig. 6, as studies discussing specific architectures are shown instead of broader reviews. Many of the Opaque

AI systems use post-hoc explanation methods (e.g. SHAP, LIME) that interpret decisions after they are made while neuro-symbolic systems integrate the explanation in the reasoning process [53]. Uncertainty in knowledge graph based and symbolic AI systems can be handled using various methods. The probabilistic approaches (e.g., factor graphs and stochastic models) quantify uncertain relationships and dependencies among entities [34], [57]. Hybrid approaches like Markov Logic Networks (MLN) combine knowledge graphs with probabilistic reasoning infer when the facts or rules are uncertain. [24]. Symbolic methods such as rough set, fuzzy logic and ontology-based methods handle incomplete and ambiguous knowledge [31], [35], [51]. Neuro-symbolic frameworks such as DFKG and DF-GRAPH reduce hallucination by grounding LLMs with structured knowledge that is extracted from knowledge graphs which improves the Opaque nature of neural AI models for forensic investigations. [6], [17].

#### *b: LIMITATIONS AND FUTURE RESEARCH DIRECTION*

Despite the benefits of these advanced hybrid and neurosymbolic methods, there are still some aspects that require improvement. A primary concern is the scalability and maintenance of semantic reasoning rules and knowledge. Many of the current investigative systems depend on manually derived SWRL rules and could benefit from automated rule learning and machine learning-based dynamic or semi-automatic rule generation to reduce the maintenance costs of semantic reasoning [11], [35]. Additionally, Challenges remain to adapt existing standards and reuse ontologies to build seamless reasoning and semantic alignment and interoperable knowledge integration between heterogeneous domains that have distinct policies, access control rules and security protocols [57]. Furthermore, the existing frameworks are often linguistic and need to further integrate multi-modal artifacts such as audio-visual evidences for semantic reasoning. [33], [37].

The utilization of graph-based reasoning and semantic technologies in neuro-symbolic architectures provides a solid mechanism to build next-generation ethical-by-design and regulation-aware AI assistants. Furthermore, Future studies may address the multi-modal gap by integrating data such as images, audio, and video alongside text and log data in semantic reasoning to achieve greater contextual awareness and adaptability for crime investigation [15], [33], [37]. There is also a clear research direction toward building more adaptive knowledge graphs that dynamically integrate evidence for transparent real-time explanations in investigations [26], [31], [53].

#### **B. RQ2: APPLICATION DOMAINS AND ADDRESSED INVESTIGATIVE PROBLEMS**

Semantic technologies are applied across diverse domains to normalize evidence and enable semantic reasoning for addressed investigative problems.

#### **1) ADDRESSED INVESTIGATIVE PROBLEMS**

##### *a: PROVENANCE TRACKING*

Arshad et al. [1] emphasize the importance of data provenance and tracking people, entities, and activities that initiate the creation of forensic data objects. Formal knowledge allows inferring, explaining, and validating subjects as initiators, practitioners, or observers from the types of objects and data evidence created by them. Furthermore, since the use of data evidence in forensics is crucial and equates to the chain of custody, a trace operator is used to manage and establish provenance tracking [49]. Fährndrich and Trollmann [39] also mention the importance of tracing evidence back to original sources for human investigators in law enforcement.

Purpose-designed ontologies enable formal mechanisms for integrity checking and facilitate chain of custody and provenance to store the origins of traces [41]. Khatiwala et al. [17] assign unique identifiers (UIDs) and SHA-256 hashes to artifacts such as device IDs, file paths, table names, and row indices for provenance, preserving end-to-end auditability and evidence consistency. Lee et al. [6] use a citation scheme and actor-aware reasoning to trace outputs that are legally defensible and interpretable for communication forensics. Saluja and Bhavsar [31] propose a six-layer cloud forensic architecture in which data provenance is a dedicated layer for digital evidence provenance tracking in knowledge graphs (e.g., PROV-O). Furthermore, they propose clock drift monitoring for future studies.

##### *b: EVIDENCE CORRELATION*

Saluja and Bhavsar [31] propose a Cloud Forensic Process Model that formalizes evidence dependencies and correlations to establish relationships between artifacts in a knowledge graph. This enables reconstruction of incident timelines or attack scenarios. Khatiwala et al. [17] note that graph algorithms can reveal critical relational structures and propose a framework to use graph-based correlations in a scalable pipeline. Liu et al. [36] build a library for evidence sequences by decomposing known attack behaviors and construct a knowledge graph to measure correlations. Jiang et al. [37] adopt a method to capture semantic relationships in KGs between multimodal data, improving retrieval. Lee et al. [6] retrieve subgraphs that maintain narrative coherence, role continuity, and causal flow for accurate forensic reasoning. Sabbu et al. [33] utilize a contextual knowledge graph to identify correlations between events and behaviors, remove ambiguity, and determine the severity of abnormalities. Peppes et al. [24] use a semantic engine to detect crime group interrelations in knowledge graphs, and SWRL rules are applied to infer new correlations. Sikos [41] also utilizes SWRL rules for data correlation stored in RDF statements from multiple data sources. Arshad et al. [49] utilize semantic queries to identify correlations and relatedness among subjects and objects across social media platforms.

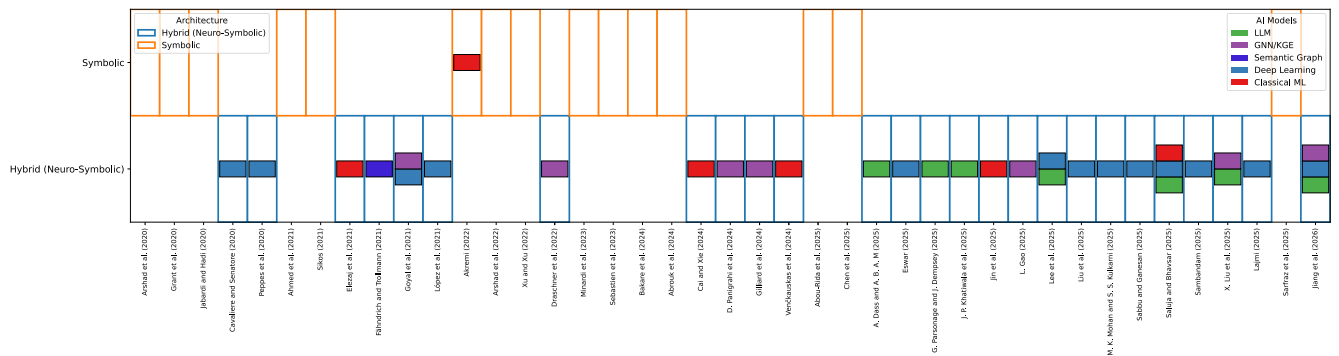


FIGURE 6. Architecture types with AI models per paper.

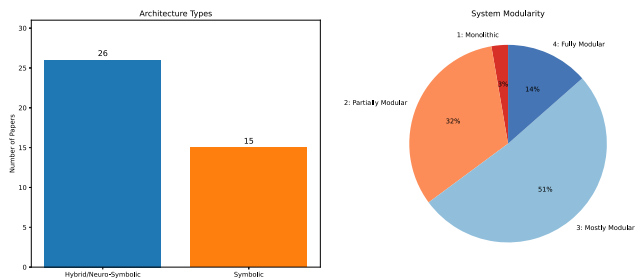


FIGURE 7. Architecture types and system modularity across papers.

c: ENTITY LINKING

Khatiwalwa et al. [17] use cross-referencing and UIDs to link entities across large datasets and correct forensic errors. Grant et al. [25] use ontologies to identify and link targets within legal frameworks. Liu et al. [36] perform entity-level linking by calculating user preference vectors and attack evidence vectors. Elezaj et al. [44] develop an ontology to store and organize components of crime evidence from multiple users across online social networks.

d: FRAUD PATTERN DETECTION

Semantic modeling is widely used to distinguish between legitimate and suspicious activities by analyzing behavioral patterns and transaction metadata.

TABLE 4. Architecture or framework per paper.

| Category          | Subcategory / Instance  | Papers (Author and Year)                                                                                                        | Freq. |
|-------------------|-------------------------|---------------------------------------------------------------------------------------------------------------------------------|-------|
| Architecture Type | Hybrid (Neuro-Symbolic) | [40] [34] [27] [26] [33] [28] [38] [36] [30] [39] [32] [37] [17] [6] [31] [54] [53] [29] [24] [56] [35] [50] [15] [7] [44] [48] | 26    |
|                   | Symbolic                | [5] [11] [45] [23] [1] [49] [43] [25] [47] [46] [42] [9] [55] [16] [41]                                                         | 15    |
| Modularity        | 1: Monolithic           | [1]                                                                                                                             | 1     |
|                   | 2: Partially Modular    | [47] [42] [27] [25] [44] [35] [40] [26] [28] [29] [9] [11]                                                                      | 12    |
|                   | 3: Mostly Modular       | [31] [45] [53] [49] [50] [17] [32] [39] [15] [38] [37] [34] [36] [30] [24] [33] [48] [16] [7]                                   | 19    |
|                   | 4: Fully Modular        | [46] [54] [23] [6] [5]                                                                                                          | 5     |

Abrouk et al. [11] develop a rule-based approach using SWRL rules related to fraud scenarios to distinguish legitimate and fraudulent transactions in the SWIFT network. Ahmed et al. [45] use ontological fraud detection mechanisms

to relate bank transactions and identify malicious behavior. X. Liu et al. [54] integrate structural modeling, entity relationships, and behavioral sequences to detect hidden abnormal patterns, logical inconsistencies, and complex fraud. Mohan and Kulkarni [30] target e-commerce fraud by detecting suspicious behavioral patterns in online transactions using evidence-based risk scores. Cai and Xie [40] introduce a two-layer graph to extract and combine semantic and syntactic fraudulent patterns in financial statements. Chen et al. [43] construct an ontology combining relationships between accounts, periods, and financial indicators to detect financial statement fraud. Gao [34] uses a hybrid graph-based approach to detect financial statement fraud. Akreimi [23] proposes a forensics-aware auto-reasoning model for detecting suspicious vehicle fraud patterns. Venčkauskas et al. [48] use ontologies to detect email-based phishing links and fraudulent activities.

Goyal et al. [38] detect fraudulent job postings using a fact-checking module trained on legitimate and fraudulent jobs. Jabardi and Hadi [47] detect online fraud and fake social media accounts using an ontology classifier. Sarfraz et al. [55] propose a fuzzy ontology that integrates behavioral patterns for effective fraudulent profile detection in online social media. López et al. [29] utilize a semantically meaningful knowledge graph to identify fraud, abuse, and anomalous healthcare billing patterns. Eswar [53] emphasize the importance of time-sensitive explainability for fraud pattern detection in neuro-symbolic AI.

e: EVENT RECONSTRUCTION

There are 13 primary studies in this review that discuss event reconstruction as a core investigative problem, addressed by semantic models that generate knowledge about forensic events. Akreimi [23] use rules to link and aggregate events for the same incidents in the Internet of Vehicles (IoV) and deduce new knowledge in real time. Arshad et al. [1] propose a knowledge-based model that reconstructs timelines of events related to incidents or criminal activities under investigation. Investigators filter data to gain insight into particular event sequences in event-based forensic

**TABLE 5. Investigative problems in studies.**

| Category             | Problem              | Papers                                                                                    | Freq. |
|----------------------|----------------------|-------------------------------------------------------------------------------------------|-------|
| Evidence Acquisition | Provenance Tracking  | [11] [49] [39] [41] [17] [6] [16]                                                         | 7     |
|                      | Evidence Correlation | [31] [17] [36] [37] [6] [33] [24] [41] [49] [46] [5] [15] [9]                             | 13    |
|                      | Entity Linking       | [17] [25] [36] [44] [26] [42]                                                             | 6     |
| Pattern Detection    | Fraud Detection      | [11] [45] [54] [30] [40] [43] [34] [23] [48] [38] [47] [29] [53] [32] [27] [28] [35] [55] | 18    |
| Reconstruction       | Event / Attack       | [23] [11] [49] [6] [31] [17] [50] [44] [39] [5] [56] [9] [16]                             | 13    |

models [49]. Lee et al. [6] use a digital forensic knowledge graph to reconstruct coherent investigative narratives. Saluja and Bhavsar [31] apply intelligent techniques for macro-level and micro-level incident timeline reconstruction, attack scenarios, and inference of adversary behavior in cloud environments. Khatiwala et al. [17] note that traditional evidence recovery methods lead to incomplete or insufficient context reconstruction for heterogeneous data such as logs, messages, and encrypted databases. Cavaliere and Senatore [50] address event reconstruction for activity tracking in neuro-symbolic architectures using the TrackPOI ontology and object identifiers to reconstruct trajectories for knowledge generation. Elezaj et al. [44] also utilize ontologies and RDF/OWL backbones for event reconstruction. Fährndrich and Trollmann [39] use marker passing in knowledge graphs to reconstruct events and relationships.

## 2) APPLICATION DOMAINS

### a: CYBERSECURITY

These papers focus on intrusion detection, threat intelligence, and protecting digital infrastructures from malicious actors. Research in this domain focuses on intrusion detection, threat intelligence, and the protection of interconnected digital infrastructures.

Abou-Rida et al. [5] focus on cybersecurity threats in connected autonomous vehicles (CAVs) that depend on interconnected systems. Checkpoint-based and forensic-aware ontologies for tracking suspicious vehicles are also addressed [23]. Cavaliere et al. [50] propose neuro-symbolic activity tracking and surveillance monitoring by detecting anomalies from various sources. Grant et al. [25] focus on Intelligence, Surveillance, Target Acquisition, and Reconnaissance (ISTAR) for offensive cyber operations and tactical cyber warfare. Panigrahi et al. [26] introduce a knowledge graph-based Intrusion Response System (IRS) using autonomous intelligent cyber-defense agents to mitigate attacks and restore infrastructure to normal operational states. Jin et al. [28] propose an advanced intrusion detection system that enhances the security of the Internet of Vehicles (IoVs). Eswar [53] discuss knowledge-driven explainable AI and knowledge graph reasoning approaches for IoT and network attack detection, counterattacks, and response. Gilliard et al. [27] explore knowledge graph reasoning for counter-network attacks. Liu et al. [36] present multi-step evidence mining for insider threats such as Advanced Persistent Threats (APTs). Elezaj et al. [44] identify crime category suspects

on social networks. Jabardi and Hadi [47] introduce semantic approaches and decision rules to detect fake social media accounts. Venčkauskas et al. [48] address email phishing risks that pose security threats to personal information. Goyal et al. [38] detect fraudulent job postings and threats in online recruitment platforms.

### b: DIGITAL FORENSICS

These studies prioritize the formalization of investigative workflows, ensure the legal admissibility of evidence, and reconstruct incident timelines in forensic life cycles.

Abou-Rida et al. [5] introduce ForenSys, a semantic assistant for cyber forensics in connected autonomous vehicles. Akremi [23] proposes a forensic-driven data model for vehicle event analysis and investigation. Ontologies play a major role in information exchange and forensic data analysis [24]. Arshad et al. [49] propose a multi-layer semantic approach for digital forensic automation in social networks. Elezaj et al. [44] present the CISMO ontology for extracting digital evidence and forensic analysis in social networks to support LEAs in developing investigative systems. They also discuss the feasibility of automated analysis operators based on formal knowledge modeling for social network forensics [1]. Cavaliere et al. [50] use ontology-based reasoning for forensic activity monitoring from diverse data sources.

Fährndrich et al. [39] use a knowledge graph-based explainable reasoning method to extract forensic evidence from confiscated data. Parsonage et al. [15] introduce an explainable neuro-symbolic digital forensic approach for criminal investigations involving digital devices. Khatiwala et al. [17] propose a knowledge graph-based evaluation of the reliability of forensic evidence discovered by LLMs. Lee et al. [6] propose a digital forensic knowledge graph-based framework for question answering over communication data such as email and instant messaging exchanges. Digital Forensic Knowledge Graphs are also used to represent, visualize, and reason over digital forensic evidence in court [16]. Saluja et al. [31] propose end-to-end automation of the forensic life cycle in cloud environments. Sébastien et al. [42] propose a framework to formalize and acquire expert knowledge using ontologies to identify drug distribution networks.

### c: FINANCE

These papers utilize semantic technologies to detect transactional anomalies, financial statement manipulation, and e-commerce fraud.

Dass et al. [32] focus on financial complaint classification and fraud detection. Ontology-based approaches to detect fraudulent transactions in the SWIFT network or to develop financial fraud deterrence models (FFD) have also been proposed [11], [45]. Sambandam [35] proposes an ontology-based expert system for detecting suspicious transactions in financial institutions. Semantic-structural co-driven fraud detection frameworks have been proposed for

**TABLE 6.** Crime investigation domain in studies (RQ2).

| Domain            | Papers (Author and Year)                                                         | Freq. |
|-------------------|----------------------------------------------------------------------------------|-------|
| Cybersecurity     | [5] [23] [50] [25] [26] [28] [53] [27] [36] [44] [47] [48] [38] [55] [37]        | 15    |
| Digital Forensics | [5] [23] [24] [49] [44] [1] [50] [39] [15] [17] [6] [31] [9] [46] [42] [16] [41] | 17    |
| Finance           | [32] [11] [45] [35] [54] [43] [40] [53] [34] [30]                                | 10    |
| Healthcare        | [53] [29]                                                                        | 2     |
| Other             | [33] [56]                                                                        | 2     |

complex financial scenarios [54]. Ontology reasoning has also been used to detect misstatement accounts in financial statements [43]. Cai et al. [40] propose an explainable two-layer knowledge graph approach for fraud detection in financial statements. Eswar [53] proposes a real-time, transparent, and adaptive framework based on neuro-symbolic AI to explain why transactions are flagged as suspicious or fraudulent. Gao [34] develops explainable AI approaches tailored for financial regulatory compliance. Knowledge graph embeddings have also been used to capture behavioral and temporal patterns in transaction data to detect financial fraud [30].

#### d: HEALTHCARE

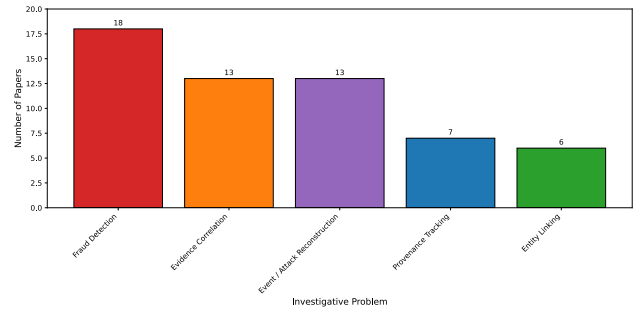
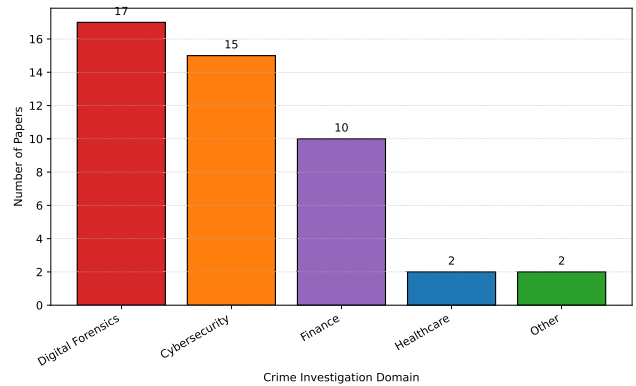
The application of semantic models in healthcare supports policy compliance and medical integrity. Eswar [53] uses neuro-symbolic integration for explainable diagnosis and medical predictions. López et al. [29] detect billing non-compliance with healthcare insurance policies.

#### e: OTHER DOMAINS

Semantic technologies are also used for crime detection in other domains. Nguyen et al. [58] utilize a blockchain-based system combined with ontologies to capture knowledge about learning activities and authenticate educational certificates. Lajmi et al. [56] use a hybrid deep learning and ontology framework for culpability assessment and explainable, legally grounded reasoning in accident resolution. Sabbu et al. [33] propose a knowledge graph-based abnormal human activity recognition method for crime analysis and public safety. Jiang et al. [37] propose an explainable knowledge graph-based violence detection surveillance method to enhance security.

### 3) EVALUATION METHODS

Table 7 demonstrates how various studies evaluated their methods for the addressed investigative problems across different domains. Thirteen primary studies use case studies that demonstrate the qualitative aspects of symbolic or hybrid systems in realistic investigations and show feasibility and practical applicability in a smaller scenarios (e.g. Luckin Coffee Financial Fraud or Galaxy Note forensic images, illegal wildlife trafficking, etc.) [16], [17], [43]. The datasets

**FIGURE 8.** Distribution of investigative problems across studies.**FIGURE 9.** Crime investigation domains in studies.

used in the case studies focus on incident-specific and single investigations that are hard to scale and generalize.

Four studies use simulation as a proof-of-concept in emerging fields such as connected autonomous vehicle(CAV) forensics where real-world incident data is limited [5]. Simulations use artificially generated logs of fraud data to simulate and verify investigative rules, semantic workflows and refine ontologies [9], [23]. However, benchmark datasets and real world experiences is required for large scale deployments.

Twelve studies use benchmarks, which uses objective an quantitative assessments to measure the performance against state-of-the-art based on standardize metrics. Public benchmark datasets such as CICIDS2017, CERT-r4.2/r5.2, CSMAR for cybersecurity and PaySim for financial benchmarking are used for reproducible comparison against established methods [5], [11], [32]. Studies have measured both predictive and explanatory performance of their proposed frameworks against multiple baseline approaches and shown the rigor and effectiveness of multi-metric comparison [6], [40].

Finally, thirteen studies use real-world experiments as shown in Fig 10. These evaluations prioritize authentic logs that reflect real-world complexities in operational systems for measuring the effectiveness of a framework. For example IEEE-CIS fraud detection dataset or AIT log dataset that contains authentic IDS logs and IISO 20022 formatted SWIFT messages private datasets are used

[11], [27], [45], [54]. The proprietary nature of some of these datasets hinders third-party audits and constrain the scientific reproducibility.

#### 4) DISCUSSION OF RQ2: APPLICATION DOMAINS AND ADDRESSED INVESTIGATIVE PROBLEMS

##### *a: KEY FINDINGS AND COMPARATIVE ANALYSIS*

In this systematic review, Fraud detection is identified as the most widely addressed investigative problem that has been discussed in 18 primary studies as shown in Fig. 8. The fraud detection studies are conducted within a variety of critical contexts such as SWIFT network transactions, financial statements, healthcare billing, and online recruitment [11], [29], [38], [40]. Among other investigative problems, Evidence correlation and event/attack reconstruction are subjects in approximately half of the primary studies [6], [31]. Furthermore, Provenance tracking is discussed in seven studies as a fundamental requirement for accurately detecting, tracking, and validating the people, entities, and objects that are involved in forensic investigations for legal admissibility [1]. Finally, Six studies address entity linking as shown in Table 5 in ontologies to facilitate organizing forensic data from various platforms [36], [44]. Digital forensics is the primary application domain, with 17 studies focusing on formalizing workflows to ensure legal admissibility of evidence and automate the forensic life cycle [1], [31], [44]. As shown in Fig. 9, 15 studies focus on the cybersecurity domain, addressing topics such as intrusion detection, advanced persistent threats (APTs), and threat intelligence, particularly in connected autonomous vehicles (CAVs), social networks, and online platforms. Ten studies are in the finance domain, where semantic technologies are used to detect anomalies, financial statement manipulation, and e-commerce fraud [32], [34], [45]. Two studies focus on identifying non-compliance and medical integrity issues using semantic technologies [29], [53]. Two additional studies address areas such as violence detection and accident resolution [37], [56]. The complete list of domains and their frequencies is presented in Table 6. The primary focus on fraud detection among the studies indicates the importance of semantic modeling to distinguish and analyze metadata and behavioral sequences to detect suspicious or fraudulent activities that commonly manifest as cybersecurity threats or financial crimes [11], [23]. The results further implies that graph structures and knowledge graphs are increasingly used to formalize dependencies and artifacts from fragmented data, which can be utilized for event reconstruction in complex crime investigation cases such as incident timelines and attack scenarios [6], [31]. The emphasis on provenance tracking and the use of a trace operator suggests the necessity to establish a legally acceptable chain of custody [1]. This ensures that the automated forensic processes are reproducible, explainable, and rigorously testable to adhere to legal constraints of judicial proceedings [1]. Furthermore, the large-scale investigative tasks such as contextual

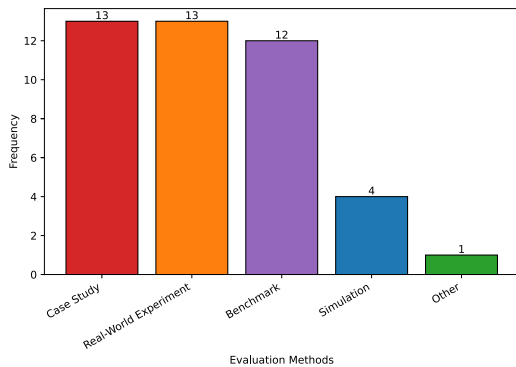
reconstruction and event correlations are used cloud-oriented approaches while device-level methods prioritize evidence integrity and provenance tracking [17], [31]. The current body of literature demonstrates that various empirical and qualitative methods are used for validation. Twelve primary studies utilize case studies to evaluate the addressed problems or to construct detailed representations of investigative phenomena [1], [24]. Furthermore, Real-world experiments and benchmarking, are frequently used to validate detection accuracy for example in SWIFT transactions or financial complaint analysis [11], [32]. Some of these evaluations argue that combining the semantic understanding of unstructured text with the structured reasoning of financial domain knowledge enables systems to outperform plain machine learning models [32]. Nevertheless, Simulation is used less frequently, with four papers that evaluate their proposed frameworks in controlled simulation environments [9], [23]. This suggests that the field prioritizes the practicality and empirical foundation of results that are validated from real data over synthetic modeling.

##### *b: LIMITATIONS AND FUTURE RESEARCH DIRECTION*

There are also technical and socio-technical considerations for the deployment of these semantic technologies. For example while some applications benefit from the ontologies there are some scalability challenges with large datasets [11]. The integration process is complicated due to the heterogeneous nature of the data from various devices, IoT, cloud, and IT systems that complicates the integration and analysis. Furthermore, the manual processes that are involved still remain time-consuming and prone to errors [5]. Additionally, There are large-scale graph processing limitations such as query response time that increase exponentially as the graph complexity grows. Furthermore, some performance bottlenecks are related to general-purpose graph query engines that should be optimized for domain specific reasonings that have complex regulatory constraints [34]. The Opaque nature of many AI applications in the judicial context is still considered a barrier since most AI models, particularly deep learning, lack transparency, context-awareness, and the ability to derive new knowledge from evidence [5]. Moreover, large language models (LLMs) in the forensic domain often generate output that is hallucinated or unverifiable [6], [31]. Finally, access restrictions, driven by privacy regulations and limited supplier cooperation is a significant challenge that hinders comprehensive investigations [5]. Future studies should focus on the automation of the forensic and analysis process to manage large data sets [1]. Provenance Tracking should be prioritized as a dedicated layer in knowledge-graph based forensic architectures. Provenance models (e.g. PROV-O) with integrity preserving, verifiable and tamper resistance blockchain-based methods can further improve traceability and legal admissibility of digital evidence in cloud environments [31]. Formal representations also can be further studied to be utilized for event reconstruction and generating incident timelines and attack scenarios [31]. The

**TABLE 7. Evaluation methods in studies (RQ2).**

| Evaluation Method     | Representative Papers (Author and Year)                         | Freq. |
|-----------------------|-----------------------------------------------------------------|-------|
| Case Study            | [1] [50] [43] [26] [44] [53] [15] [17] [46] [24] [42] [41] [16] | 13    |
| Simulation            | [45] [23] [49] [9]                                              | 4     |
| Benchmark             | [40] [37] [28] [34] [6] [36] [30] [33] [35] [48] [45] [55]      | 12    |
| Real-World Experiment | [5] [32] [11] [27] [38] [47] [29] [31] [54] [56] [55] [25] [34] | 13    |
| Other                 | [39]                                                            | 1     |



**FIGURE 10. Evaluation methods in studies.**

studies that are focused on fraud detection can benefit from dynamic rule generation and use advanced analysis operators based on domain correlations [1], [11]. Furthermore, standardizes and extensible taxonomies can be used in applications such connected autonomous vehicles (CAVs) and social networks for consistent classification [31]. One of the major challenges in the future research is to automatically generate structured data from hybrid data efficiently while considering operating system differences and standard semistructured formats on IoT devices [41]. Finally, automated investigation systems should ground the outputs of large language models (LLMs) in formally defined forensic ontologies to reduce hallucinations and enhance trustworthiness. [31].

### C. RQ3: UNDEREXPLORED SOCIO-TECHNICAL AND RESPONSIBLE AI ASPECTS

Semantic technologies and knowledge graphs provide mechanisms to implement technical solutions that capture formal knowledge and enable semantic reasoning. However, it is essential to understand their impact on various socio-technical and responsible AI aspects.

#### 1) ACCOUNTABILITY

Accountability is defined as the possibility of determining who is responsible for a decision or output [52]. Accountability issues stem from biases in outputs, inconsistent refinements, and the lack of structured validation for AI-generated outputs [17].

#### 2) EXPLAINABILITY AND TRANSPARENCY

Forensic investigations require the interpretation and tracing of digital evidence to support reasonable and legally defensible conclusions. However, most AI models are Opaque systems and lack transparency, context awareness, and the ability to incorporate or derive new knowledge from evidence for explainable and fact-based analysis, making it difficult for users to understand the reasons behind model decisions [5], [37]. Another challenge is avoiding hallucinatory or unverifiable content and obscure reasoning paths, which are critical limitations of many generative AI models. Therefore, it is necessary to provide traceable and structured reasoning chains linked to original sources and aligned with investigative logic, allowing LEA agents to verify and understand outcomes.

Marker passing is an explainable method used during the construction or utilization of knowledge graphs [6], [31], [39]. Graph-based methods can also employ multi-hop reasoning algorithms to traverse complex relationships while maintaining interpretability [34]. Jabardi et al. [47] propose an ontology-based classifier that provides interpretable decision rules for investigators.

In many cases, machine learning models exhibit poor interpretability [43]. Rule-based reasoning is applied to ontologies to infer implicit facts not explicitly described, thereby increasing solution reliability [45]. When relationships between domain-specific entities are not considered, models may fail to identify facts correctly and transparently, reducing system trustworthiness and negatively affecting user experience [38]. Explainable artificial intelligence (XAI) methods have been developed to improve interpretability and support more transparent decision-making [35]. Adaptive real-time symbolic reasoning and explanations in neuro-symbolic AI systems further improve transparency and reliability [53]. Jin et al. [28] use SHapley Additive exPlanations (SHAP) to clarify feature influence on model decisions. Conclusions drawn from automated forensic processes should be legally explainable using formal theories and knowledge models to ensure admissibility in courts of law [1], [49]. Hybrid approaches integrating domain-specific knowledge into AI models can enhance transparency, interpretability, accuracy, and effectiveness [32].

#### 3) GOVERNANCE AND POLICY COMPLIANCE

Integrated forensic requirements and implemented rules vary across countries, and cyber-forensic or defense operations must be adapted to national and international laws [23], [25]. Regulatory compliance audits require information ingestion channels to detect, respond to, and report evolving threats [26]. Neural models often lack key forensic principles requiring investigative actions to be auditable and reproducible by third parties [15]. Identifying policy non-compliance is frequently narrow in scope and complex or manual. Explainable AI techniques can be tailored to meet regulatory compliance requirements [29], [34]. Malicious

cyber activities often ignore political boundaries and legal jurisdictions [24]. Modern crime investigation frameworks have expanded legal admissibility standards to address emerging technologies within judicial contexts [6].

#### 4) FAIRNESS AND BIAS MITIGATION

AI tools can assist investigators and reduce biases in traditional manual crime detection and forensic processes [42]. Adaptive real-time explanations and human oversight improve fairness and significantly reduce bias [29], [53]. Contextual bias in generative AI models, such as LLMs, arises from co-occurrence patterns and explicit structural relationships; incorporating structured knowledge can mitigate these biases [17]. Mohan and Kulkarni [30] emphasize balancing business value with ethical principles related to fairness and transparency.

#### 5) RISK, SAFETY, AND TRUST

Many modern deep learning models make decisions that are difficult to interpret, leading to trust concerns, particularly in high-stakes and critical applications [53]. Over-surveillance can also negatively impact trust [36]. Minardi et al. [46] use criminal risk ontologies and qualitative risk mini-models to represent socio-technical risks. Semantically enriched data and contextual knowledge graphs can help diagnose harmful behavior, prevent crime, and enhance public safety [33].

#### 6) PRIVACY, ETHICS, AND HUMAN-CENTRIC CONSIDERATIONS

Abrouk et al. [11] suggest that future studies uphold strict ethical and privacy standards to protect sensitive data used within ontologies. Human review processes should be refined to minimize false positives and protect individuals from unwarranted scrutiny and over-surveillance [36]. Human understanding, oversight, and control are considered first-class concerns in AI design, reducing bias and policy errors [29]. In legal and judicial contexts, privacy considerations for individuals involved in legal cases are essential, although privacy compliance requirements may restrict comprehensive investigations [5], [44]. LEAs are expected to minimize unnecessary intrusion into subjects' privacy [15]. Automated attack recognition methods must consider privacy and ethical implications during network monitoring and personal data analysis [27]. Forensic taxonomies support the investigation of privacy risks by describing smart objects and vulnerabilities [41].

Some studies address socio-cognitive aspects in cybersecurity ontologies; however, biometric ontologies for identifying individuals based on physical characteristics such as fingerprints, iris patterns, or facial features remain underdeveloped [44]. LEAs should also apply anonymization techniques during data seizure to prevent privacy breaches while preserving evidence authenticity and legal admissibility [23]. Cavaliere et al. [50] use multi-agent systems with evaluator agents executing SPARQL queries to provide decision support to human operators. Cyber Threat Intelligence (CTI)

sharing frameworks that use domain-specific ontologies and semantic parsers enable information exchange while preserving privacy and interoperability [48]. A trade-off between transparency and the risk of sensitive data disclosure must be considered [7]. Dass et al. [32] propose extending LLM-knowledge graph integration to edge environments to support real-time, privacy-preserving fraud analysis.

#### 7) DISCUSSION OF RQ3: SOCIO-TECHNICAL AND RESPONSIBLE AI ASPECTS

The synthesis of evidence for RQ3 examines socio-technical and responsible AI aspects of knowledge graphs and semantic technologies [1], [6], [51]. These findings are further discussed within the context of responsible AI governance for crime investigation and digital forensics.

##### *a: KEY FINDINGS AND COMPARATIVE ANALYSIS*

The systematic review reveals that among various aspects explainability and transparency are the most frequently addressed aspects which are discussed in 19 studies. These studies emphasize that semantic technologies support legal admissibility in forensic settings by enabling explainable and structured reasoning that aid human investigators to understand AI decision-making processes [1]. Five studies address fairness and bias reduction that highlight the necessity for adaptable, real-time explainable AI methods to mitigate discriminatory outcomes [17], [42], [53]. Additionally, privacy, ethics, and human-centric considerations are emphasized in 13 studies as shown in Fig. 11, which demonstrate the critical focus on the data protection and privacy preserving methods for crime investigation that use open evidence in knowledge-graph based AI and semantic reasoning [7], [11]. Conversely, accountability remains underexplored, with only three studies addressing the determination of responsibility for AI-generated decisions [17], [52].

The RQ3 findings suggest that semantic technologies and knowledge graphs provide mechanisms to improve transparency and interpretability by capturing formal knowledge to enable semantic reasoning [1], [6]. The studies use transparent semantic rule-based logic and structured knowledge graphs to audit Opaque AI systems with fact-based reasoning chains that aligns investigative conclusions with digital evidence in neuro-symbolic architectures [5], [6]. In many high-risk environments such as healthcare and smart-city surveillance, these technologies provide logically grounded alerts and risk assessments that improve transparency and explainability [46], [51], [53]. Frameworks such as NS-XAI adopt adaptive real-time symbolic reasoning for bias reduction but still recommend human-in-the-loop to further mitigate biases or policy errors. The integration of Large Language Models (LLMs) with Knowledge Graphs (KGs) also help ground neural outputs with structured knowledge and ensures that semantic reasoning frameworks remain legally defensible and adaptable to forensic governance requirements [23], [26], [41]. The judicial requirements for forensic processes can be captured in unified frameworks

(e.g. KDCFI) that enable structured reasoning as conceptual metadata to ensure evidence is compliant with standards such as NISTIR 8006 and acceptable for legal automation [31].

The Opaque deep learning methods that have been developed lack the rigor that is required for judicial use cases and therefore, semantic technologies support legal admissibility in forensic settings by enabling explainable and structured reasoning [1], [6]. The semantic approaches study various aspects of responsible AI such as explainability, fairness, accountability, etc. through structural or logical lens rather than a statistical learning model. The relevant socio-technical categories and associated studies are listed in Table 8.

#### b: LIMITATIONS AND FUTURE RESEARCH DIRECTION

Despite the advances in semantic modeling there are still significant ethical limitations to address: High transparency requirements in semantic technologies necessitate a careful analysis of how privacy is impacted since there is a trade-off between transparency and privacy-related risks [7]. Furthermore, human-in-the-loop(HITL) control mechanisms should be designed to ensure compliance with privacy laws. However, human oversight and review processes should be refined to protect individuals from over surveillance or private information breaches and data-disclosures [7], [11]. Furthermore, lack of structured validation, inconsistent refinements and reasoning biases in the AI-generated output can lead to accountability issues and complicate responsibility determinations [17]. The lack of focus on the accountability is a gap for research in this field. Accountability is considered as the possibility of determining who is responsible for a decision or output [52]. Furthermore, future research should explore bias reduction and the use of adaptable, real-time explainable AI methods to mitigate discriminatory outcomes [42], [53]. legally defensible semantic frameworks that are adaptable for forensic governance should be developed. [23], [41]. The hybrid deployment approaches between cloud and self-hosted LLMs on the edge perform that regular tasks are suggested for data-sovereignty, chain-of-custody compliance and cost effectiveness [6]. Furthermore, Human-in-the-loop controls should be introduced to manage sensitive personal while disclosing the information for investigative transparency [7], [11].

#### IV. THREATS TO VALIDITY

In accordance with established methodological standards for secondary research in software engineering and forensic computing, this systematic literature review (SLR) accounts for potential threats to validity through a categorized mitigation strategy [59]. These measures are designed to ensure the integrity of the scientific foundation and the reproducibility of the synthesized findings.

##### A. CONSTRUCT VALIDITY

Construct validity addresses the degree to which the operational measures employed in the SLR accurately reflect

TABLE 8. Socio-technical and RAI challenges.

| Category                                         | Papers                                                                                     | Freq. |
|--------------------------------------------------|--------------------------------------------------------------------------------------------|-------|
| Accountability                                   | [32] [52] [17]                                                                             | 3     |
| Explainability / Transparency                    | [5] [37] [39] [6] [31] [34] [47] [43] [45] [38] [35] [53] [28] [1] [49] [32] [40] [9] [54] | 19    |
| Governance / Compliance                          | [23] [25] [26] [15] [34] [29] [24] [6]                                                     | 8     |
| Fairness / Bias Mitigation                       | [42] [53] [29] [17] [30]                                                                   | 5     |
| Risk, Safety, & Trust                            | [53] [36] [46] [33]                                                                        | 4     |
| Privacy, Ethics and Human-Centric Considerations | [11] [36] [29] [44] [5] [15] [27] [41] [23] [50] [48] [7] [32]                             | 13    |

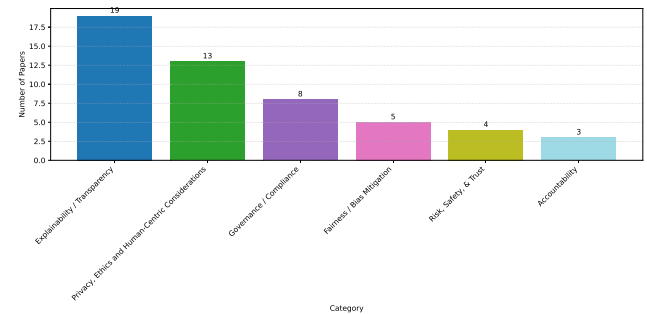


FIGURE 11. Number of papers per Socio-technical / RAI challenge category.

the theoretical concepts under investigation, specifically the intersection of graph-based knowledge representation (KR) and cybercrime investigation [59]. This risk was mitigated through the systematic design of search queries utilizing standardized, controlled keywords such as “ontology,” “knowledge graph,” “RDF/OWL/SWRL,” and “neuro-symbolic.” Furthermore, the inclusion of domain-specific terminology related to forensics and fraud detection limits the search scope to remain focused on the pertinent intersection of technologies and investigative applications. The search queries were adapted based on each database’s specific syntax requirements (e.g., Scopus vs. IEEE Xplore) to extract the relevant studies from disparate platforms. Additionally, the concrete search string for each database and the search dates for queries saved and made available online to allow the refreshment of the primary studies and reproduction of the review studies. To capture influential studies that may utilize non-standardized synonyms or keyword combinations, one iteration of backward snowballing was applied to all studies that passed the quality assessment criteria.

##### B. INTERNAL VALIDITY

Internal validity concerns the rigor of the data extraction process and the mitigation of researcher bias during the screening and synthesis phases [59]. The review protocol was subjected to systematic oversight, with screening, data extraction, and architectural classification decisions documented and rigorously revised at each milestone to receive validation from a primary supervisor and a supporting

**TABLE 9.** Database search queries used in the systematic literature review.

| Database                         | Search Query                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scopus                           | (TITLE-ABS-KEY ( semantic* OR symbolic* OR neuro*symbolic* ) AND TITLE-ABS-KEY ( "knowledge graph*" OR "knowledge-graph*" OR KG OR SPARQL OR RDF OR OWL OR "linked data" OR ontolog* OR SHACL OR SWRL) AND TITLE-ABS-KEY ( crim* OR fraud* OR forensic* OR cyber*crim* ) AND TITLE-ABS-KEY (transpare* OR explaina* OR ethic* OR moral* OR responsibl* OR socio* OR policy* OR bias* OR human* OR value-cent* OR value-sensi* OR fair* OR accountab* OR interpret* OR governance OR audit* OR regulat* OR safe* OR trust* OR risk* OR threat* OR privacy OR dignity OR agency OR autonomy))                                                                                                                                                                                                                                                    |
| IEEE Xplore                      | (semantic* OR symbolic* OR neuro-symbolic*) AND ("knowledge graph*" OR "knowledge-graph*" OR KG OR SPARQL OR RDF OR OWL OR "linked data" OR ontolog* OR SHACL OR SWRL) AND (crim* OR fraud* OR forensic* OR cybercrim*) AND (transparency OR transparent OR explainability OR explainable OR ethics OR ethical OR ethic OR moral OR morality OR responsible OR responsibility OR socio* OR policy OR policy-based OR bias OR biased OR humanized OR human-oriented OR human-centered OR value-centered OR value-sensitive OR fair OR fairness OR accountability OR accountable OR interpret* OR governance OR audit OR audits OR auditable OR regulate OR regulation OR regulatory OR safety OR risk OR risk-management OR "risk assesment" OR risks OR trust OR trustworthy OR threat OR threats OR privacy OR dignity OR agency OR autonomy) |
| Web of Science (Core Collection) | TS=(semantic* OR symbolic* OR neuro-symbolic*) AND TS=("knowledge graph*" OR "knowledge-graph*" OR KG OR SPARQL OR RDF OR OWL OR "linked data" OR ontolog* OR SHACL OR SWRL) AND TS=(crim* OR fraud* OR forensic* OR cybercrim*) AND TS=(transpare* OR explaina* OR ethic* OR moral* OR responsibl* OR socio* OR policy* OR bias* OR human* OR value-cent* OR value-sensi* OR fair* OR accountab* OR interpret* OR governance OR audit* OR regulat* OR safety OR trust* OR risk* OR threat* OR privacy OR dignity OR agency OR autonomy)                                                                                                                                                                                                                                                                                                       |
| ACM Digital Library              | ( title:( semantic* OR symbolic* OR "neuro*symbolic*") AND ("knowledge graph*" OR "knowledge-graph*" OR KG OR SPARQL OR RDF OR OWL OR "linked data" OR ontolog* OR SHACL OR SWRL) AND (crim* OR fraud* OR forensic* OR "cyber*crim*") AND (transpare* OR explaina* OR ethic* OR moral* OR responsibl* OR socio* OR policy* OR bias* OR human* OR value-cent* OR value-sensi* OR fair* OR accountab* OR interpret* OR governance OR audit* OR regulat* OR safe* OR trust* OR risk* OR threat* OR privacy OR dignity OR agency OR autonomy) ) OR abstract:( same search structure as title ) OR keywords:( same search structure as title ) )                                                                                                                                                                                                    |
| ScienceDirect                    | ( semantic OR symbolic OR neuro-symbolic) AND (knowledge) AND (crime OR fraud OR forensics) AND (explainable)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| SpringerLink                     | ("semantic AI" OR "semantic reasoning" OR "symbolic reasoning" OR "neuro-symbolic") AND ("knowledge-based") AND (crime OR fraud OR forensic) AND (explainable)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

faculty member. Also professional advice from the university librarian was incorporated to ensure the breadth and coverage of the search from credible sources. The use of the Covidence platform provided a standardized tools and environment for de-duplication and systematic screening of 597 candidate papers, ensuring a traceable and transparent selection process. Moreover, the application of formalized quality assessment criteria (QC), scored on a 0–2 scale and ensuring total quality scores is greater than 4, grounds the final synthesis in technically sound and empirically validated research rather than purely speculative conceptualizations.

was used as the primary tool for systematic screening and de-duplication of the 597 candidate papers, providing a traceable selection audit trail

### C. EXTERNAL VALIDITY

External validity defines the extent to which the synthesized results can be generalized to the broader domain of cybercrime investigation and semantic technologies [59]. A primary limitation to the external validity of this review is its exclusive focus on peer-reviewed English-language literature published between 2020 and 2026. While this temporal window was selected to capture the maturation of Responsible AI standards and current neuro-symbolic trends, it may exclude foundational ontological research published prior to 2020 or relevant findings from practitioner-driven reports, industrial deployments, and patented proprietary tools that fall outside academic indexing scopes. Language bias is also a factor, as relevant studies published in languages other than English were not retrieved or analyzed.

Consequently, while the review provides a scientifically robust overview of current academic trends, it may not fully encapsulate the industrial and regulatory landscape.

### V. CONCLUSION

This systematic literature review synthesizes the current landscape of graph-based knowledge representation (KR) and semantic technologies within the multidisciplinary domains of digital forensics, fraud detection, and cybercrime. The systematic research steps adopt the Kitchenham methodology to examine how intelligent conceptual frameworks, applications, and socio-technical aspects of semantic technologies are studied to provide a scientific foundation for legally admissible digital investigations [1].

This study makes several key contributions. The analysis of architectural trends (RQ1) reveals a definitive paradigm shift from traditional, isolated forensic tools toward intelligent forensics characterized by layered and hybrid neuro-symbolic architectures [15], [49]. The integration of Large Language Models (LLMs) with structured Knowledge Graphs (KGs) via Knowledge-Augmented Generation (KAG) is shown to effectively mitigate generative hallucinations. This hybrid approach grounds AI-driven decisions in ontology-backed evidence reasoning and knowledge inference [6], [31]. The technical implementation of these systems uses OWL and RDF, which provide machine-interpretable and trackable representations necessary for handling diverse and heterogeneous data sources in computing environments [5], [41]. Practitioners can implement KAG in agentic workflows to adopt neural models for initial anomaly detection while

routing high-stake investigative decisions to the semantic layer for formal verification [52]. The shift toward intelligent forensic systems with unified graph-based semantic knowledge, such as DF-GRAPH, provides a glass-box mechanism with significant value for traceable and accurate legal reasoning in crime investigation [5], [11], [32]. However, challenges remain in scaling these graph-based reasoning solutions to enterprise-level platforms with large datasets (e.g., 8 TB+) that support cross-jurisdictional interoperability without latency issues. Furthermore, there is a lack of universally accepted standards for building top-level forensic ontologies [34], [41]. Future work should focus on standardizing structured knowledge representation artifacts for cybersecurity and forensic ecosystems and adopting formats such as CASE (Cyber-investigation Analysis Standard Expression), UCO (Unified Cyber Ontology), and STIX (Structured Threat Information eXpression) to enable global interoperability among forensic tools [17], [41]. Additionally, it is essential to invest in methods that apply privacy-preserving federated learning over decentralized data and support anonymization of personal information in ontologies in accordance with regulations such as GDPR [32], [36].

Regarding application domains and investigative challenges (RQ2), the review confirms that high-stakes domains such as connected and autonomous vehicles (CAVs), the SWIFT financial network, and online social media forensics successfully integrate and operationalize semantic technologies [11], [23], [44]. These technologies address critical forensic and investigative problems, including provenance tracking for maintaining the chain of custody, multi-step evidence correlation for insider threat detection, and automated reconstruction of crime scene timelines [16], [36], [49]. Furthermore, LEA investigators can use multi-modal heterogeneous evidence in a unified knowledge graph to discover semantic correlations and reconstruct events from disparate data sources [37].

Finally, the investigation of socio-technical and responsible AI aspects (RQ3) identifies several underexplored areas. Formal knowledge representation provides interpretable structures and human-readable decision rules for use in Law Enforcement Agencies (LEAs); however, the reliability, accountability, governance of knowledge structures, as well as privacy and bias mitigation, remain major challenges for AI models [7], [47]. Furthermore, as investigations increasingly involve the Social Internet of Things (SIoT) and complex cloud infrastructures, there is a need to develop lightweight trust models that enhance privacy in social contexts [31], [57]. Researchers should also develop custom metrics for rigorous forensic requirements of the court which measure the integrity, credibility and admissibility aspects of semantic technologies and KG-based investigation systems beside the machine-learning metrics (e.g. F1-Score, Accuracy) [17].

Researchers should prioritize the development of standardized Forensic Soundness Metrics, such as Evidence Extraction Accuracy (EEA), Knowledge Graph

Connectivity Accuracy (KGCA), and Chain-of-Custody Adherence (CCA). These should be validated against standardized ground-truth datasets—such as the Cellebrite Capture-the-Flag (CTF) challenges—to provide a statistically defensible basis for AI-driven testimony in court

In conclusion, graph-based semantic technologies offer a robust path toward automated forensic analysis that is more transparent, auditable, and legally admissible. Future research should prioritize the development of cross-domain and adaptive ontologies capable of integrating evidence across platforms. In addition, hybrid models should be refined to support human-in-the-loop explainability and validation of decision rules. These efforts will help establish a more trustworthy and legally admissible paradigm for intelligent and explainable crime investigation.

## APPENDIX

### DATABASE SEARCH QUERIES

The search logic for each search engine is implemented as shown in Table 9.

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